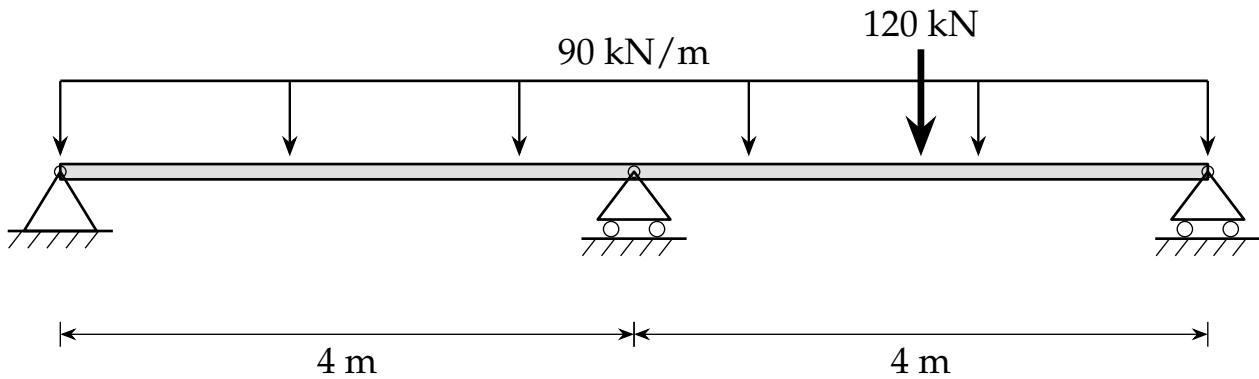


ME3180 Finite Element Method: Project Problems
If any data is missing, assume a meaningful value and justify it.

GROUP 7



Given: $E = 210 \text{ GPa}$, $I = 100 \times 10^{-6} \text{ m}^4$.

1. Write the differential equation.
2. Derive the corresponding weak form.
3. Derive the elemental stiffness and force vector for one element.
4. Write a finite-element code to solve the problem.
5. Plot the profile of deflection, shear force, and bending moment.
6. Show the change in maximum deflection as a function of the total number of elements.
7. Show the change in maximum deflection as a function of the higher-order approximation.